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(1) Research Program/Project/Study Title:	
	Teaching Physical Education during the Pandemic: The CapSU Roxas City Experience
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz
Region:	Region VI

(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization - Physical Education
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(9A) Abstract

Teaching Physical Education was commonly held outdoors and was socially engaging since there are no restrictions like having close contact with others. The shift to online instruction caused by the pandemic presented numerous challenges for PE teachers. This study aimed to collect and compile teaching approaches, methods and strategies that were used in teaching PE during pandemic. Respondents were asked for their insights and documentations while teaching the physical education class during pandemic and were given questionnaires whereas the questions are in relation to their teaching strategies, methods, approaches and modalities that are implemented and the significant impact of teaching physical education. This study utilized a descriptive research design utilizing eleven PE Teachers of the CapSU Roxas City Main Campus. Thematic analysis by Braun and Clarke was used in generating themes of the data. The modes of teaching utilized were as follows: modular learning and multimedia applications - facebook group page, facebook messenger, google meet, google classroom and youtube. Constructivist approach, child-centered method, and lecture and demonstration strategies were the approaches, methods and strategies were used by PE teachers in teaching Physical Education during the pandemic respectively. Pandemic has had a significant impact on PE teachers, who have faced a multitude of challenges and uncertainties in adapting to new teaching methods and adjustments. This study benefits the curriculum planner, teachers, students, parents and future researchers.

(9B) Keywords

Teaching Physical Education, pandemic, approaches, methods, strategies, magazine

(10) Introduction

COVID-19 is a respiratory disease caused by the SARS-CoV-2 virus, became a global pandemic in spring 2020, forcing schools around the world to change their instructional methods. The shift to online instruction caused by the pandemic presented numerous challenges for PE teachers (Garcia and Weiss 2020). School closures caused by the COVID-19 pandemic reduced opportunities for youth to be physically active, affecting health disparities in this population disproportionately. Physical education is the most widely used intervention to promote physical activity among school-aged children, but teachers' perspectives on how to maintain quality programming during virtual learning periods are largely unexplored. Using a diversity, equity, and inclusion framework.

Physical education classes are important for many students in terms of getting active and establishing healthy routines. However, when schools closed, these classes were among the most difficult to adapt to an online environment. Space, weather, social distance, and other factors continue to limit what physical education teachers can do in-person and with students who learn at home. Physical Education teachers must use strategies and creativity to overcome the challenges that come with implementing online learning. PE teachers must select appropriate media to make learning more engaging. During the covid-19 pandemic, PE teachers were forced to be more open to digital literacy, as well as more creative in their learning and innovative abilities. There is a lot of hope that the solution of social, economic, and technical teaching of physical education for the learning process can be found in accordance with the educational goals of national education.

This study aimed to collect and compile documented teaching approaches, methods and strategies that were used in teaching PE during pandemic. Specific objectives were: (1) What are the modes of teaching being utilized in teaching physical education during pandemic?; (2) What are the approaches, methods and strategies being utilized in teaching physical education?; and (3) What is the perceived effect of teaching physical education during the pandemic in the performance of teachers?

(11) Materials and Methods or Methodology

The research design used in this study was simple qualitative research design, the descriptive type of research to obtain information concerning the current status of the phenomena to describe "what

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exist” with respect to variables or conditions in a situation (Key, 1997).

It is crucial to define the research instrument in this study. In addition according to Arikunto (2010) in order to facilitate the researcher, research instrument is chosen as assisting tool to collect the data. Ary (2010) stated that the most common research instruments used in qualitative research are observation, interview, and document analysis. In this study, the researcher conducted an interview to collect the data as well as pictures from the teachers. Thematic analysis by Braun and Clarke (2006) was used in analyzing the data.

The study utilized purposive sampling technique in selecting the respondents of this study to be included. The researcher chose to interview the eleven (11) Physical Education Teachers of Capiz State University, Main Campus in support to the output of the research study - a magazine in which the documentary photographs are being interpreted based on the gestures showed. It is a selective sampling method which researchers rely on their own judgment and choose the entire members of the population to participate in the data gathering procedure.

(12) Results and Discussions

MODES OF TEACHING PHYSICAL EDUCATION DURING PANDEMIC

Physical education provides the largest intervention to support the physical activity of school-aged youth, but teachers’ opinions about how to maintain quality programming during virtual learning periods remain unexplored. Although online resources are already available for educators, they are not sufficient to meet current physical education teacher needs (D’Agostino, Urte et. al 2021). Due to the COVID-19 outbreak, many countries around the world have imposed various restrictions on social life and education to slow the spread of COVID-19. These include introducing varying degrees of social isolation and restrictions on things such as social gatherings, travel, sports, leisure activities, going to work and school. Different teachers used different forms such as online classes with livestreaming, presentation, recorded videos, tasks for students, projects or just links to follow (Filiz and Konukman, 2020).

Based on the conducted interview to the physical education teachers of the Capiz State University Main Campus, multimedia and modular learning were the modes of teaching utilized in teaching physical education that conforms to the e-learning theory which built on cognitive science principles that demonstrate how the use and design of educational technology can enhance effective learning (David, 2015). Multimedia learning is one specific principle of e-learning theory, and it contends that deeper learning can be promoted using two formats among audio, visual, and text instead of one or three (Mayer, Sweller & Moreno, 2015).

Modular Learning is one of the modes of teaching that the teachers had executed for the purpose of catering the needs and learning of students despite the distance barrier caused by the pandemic. Modules were reproduced that served as a learning materials for the students to study, and learn from.

Facebook Group Page is one of the features of facebook application - an online platform that that allows people to communicate and share their common interest. Facebook Pages are places on Facebook where artists, public figures, businesses, brands, organizations and nonprofits can connect with other people or to their intended person. In this mode of teaching, students were asked to like and follow the facebook page on facebook, so that they can start seeing updates from that Page in their Feed. This result conforms with the study of Insorio and Olivares, 2021 which states that Facebook groups as alternative learning management systems (LMS) which were convenient and interactive for learning. As mentioned by Key Informant No. 1, “I utilized facebook group page, messenger and google meet in holding virtual class.”

Facebook Messenger is a proprietary instant messaging app and platform developed by Meta Platforms. Originally developed as Facebook Chat in 2008, the company revamped its messaging service in 2010, released standalone iOS and Android apps in 2011, and released standalone Facebook Portal hardware for Messenger calling in 2018. Messenger itself is a mobile application that enables chat, voice and video communications between the social media site's web-based messaging

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and smartphones. As according to Key Informant No. 5, "I use youtube, facebook messenger and facebook in presenting videos on different skills."

Google Meet is a video-communication service developed by Google. It is one of two applications that constitute the replacement for Google Hangouts, the other being Google Chat. This google meet serve as an virtual meeting to both students and teachers for the discussion of the lesson as well as for teaching demonstration for a particular skills to obtain the competencies of the a certain subject.

Google classroom is a suite of online tools that allows teachers to set assignments, have work submitted by students, to mark, and to return graded papers. It was created as a way to get eliminate paper in classes and to make digital learning possible. It was initially planned for use with laptops in schools, such as Chromebooks, in order to allow the teacher and students to more efficiently share information and assignments.

YouTube is an American online video sharing and social media platform headquartered in San Bruno, California. It is owned by Google, and is the second most visited website, after Google Search. This video application was utilized to upload the video performances of the students in their some subject area because this platform does not required a duration time. Videos were posted privately then students will copy the link of the video and sent to their respective group chat for checking and verification of the teachers. Feedback was also provided in the comment box of such application, hence, students can assess their performance through the feedback given based on their performance.

These results conform with the study of Zetou, et al. in 2022 which claims that the usability of an educational software is an important criterion for its quality and selection to be used to support learning. Thus, it is implied that multimedia has proven to be very effective in the learning process. This approach is very attractive for students to involve with, enjoy and actively interact with the content resulting in an easier and better learning.

TABLE 1. MODES OF TEACHING PHYSICAL EDUCATION DURING PANDEMIC

Participant	Responses	Clustered Theme	Emergent Theme
Informant No. 1 and 5	"I utilized facebook group page, messenger and google meet in holding virtual class."	- Facebook Group Page - Facebook Messenger - Google Meet	➤ Modular ➤ Multimedia Applications
Informant No. 2	"I use youtube, facebook messenger and facebook in presenting videos on different skills"	- Modules - Google Classroom - Youtube	
Informant No. 3 and 10	"I've utilized exploratory teaching , when the students constructs the ideas and are facilitated by the teacher during synchronous learning. Collaborative discussion was also utilized during pandemic."		
Informant No. 3 and 10	"I utilized messenger, google meet, google		

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classroom and modules.”

“I am using google
classroom accessible to
students”
Informat No. “Blended Learning and
4, 6 and 9 Hybrid”

“Modular learning and
blended”
“Distribution of modules
and blended learning”
Informat No. 8 “Modular and Blended
Learning”

Informat No. “Blended Learning and
7 and 11 Modular Learning”

“Learning using modules
and blended approach”

APPROACHES, METHODS AND STRATEGIES BEING UTILIZED IN INTEGRATING KNOWLEDGE IN TEACHING PHYSICAL EDUCATION

Approaches, methods and strategies being utilized in teaching physical education assessed the learners and teachers to cope up the learnings during pandemic. Although online resources are already available for educators, they are not sufficient to meet current physical education teacher needs (D’Agostino et al., 2021). Different teachers used different forms such as online classes with livestreaming, recorded videos, tasks for students, projects or just links to follow (Filiz and Konukman, 2020).

Constructivist approach. For teaching approaches, most of the PE teachers of Capiz State University Roxas City – Main Campus used constructivist approach of teaching. They allow their learners to construct ideas and knowledge based on their experiences and prior knowledge related to the given lessons at hand. According to Key Informant No. 5, “Instructions and modules are sent in messenger and students are encouraged to create their own ideas according to their experiences and stocked knowledge.” This finding conforms with the study of Ayaz and Sekerci, 2015 which claims that constructivist learning approach makes more significant contribution to learners’ academic achievement than traditional learning methods. The result implied that in teaching Physical Education, using the constructivist approach would make the teaching and learning process more active, relateable, and can enhance the metacognition of the students.

Child-centered method. For teaching methods, results revealed that majority of the PE teachers are utilizing child-centered method or student-centered method of teaching. As to the statement of Key Informant No. 3, “The focus of our instruction is our students for them to learn independently”. This finding conforms with the study of Andiema in 2016 which claims that teachers’ incessant utilisation of child-centred teaching methods affected the teaching and learning of psychomotor and creative activities in public schools. This result implies that student-centered method can definitely enhance students learning in a sense that they are given the equal opportunities and freedom to learn independently and meaningfully.

Lecture and demonstration. Lastly, for teaching strategies, results showed that lecture and demonstration were the most commonly used teaching strategies of the Physical Education Teachers of

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Capiz State University – Main Campus. This finding conforms with the study of Hussain in June 2020 which claims that demonstration is an effective method to improve the children's' understanding regarding abstract concepts. It enhances their proficiency, competency, receptivity, and confidence. In addition to the study of Cerbin, 2018 which claims that Lecture is a primary mode of instruction in higher education. Learning from lectures depends on deep cognitive engagement with the subject matter before, during, and after the actual presentation. The result implied that in teaching Physical Education, lecture and demonstration should come together for the better learning of students. Thus, physical education teaching must have the knowledge and skills in teaching competencies to students in relation to physical education.

TABLE 2. APPROACHES, METHODS AND STRATEGIES BEING UTILIZED IN INTEGRATING KNOWLEDGE IN TEACHING PHYSICAL EDUCATION

Participant	Responses	Clustered Theme	Emergent Theme
Informat No. 5 and 8	"Instructions and modules are sent in messenger and students are encouraged to create their own ideas according to their experiences and stocked knowledge."	• constructivist approach • child-centered method • lecture • online presentations • demonstration	➤ Approaches ➤ Methods ➤ Strategies
Informat No. 2	"I allow my students to generate ideas from their existing knowledge"		
Informat No. 1 and 9	"Demonstration is the best way to integrate knowledge and skills into teaching physical education. With this, students can improved their mastery of the skills, memorize steps, and solve problems."		
Informat No. 10 and 7	"Giving link and execute the skills that they are watching in youtube."		
	"Demonstrations of the skills first"		
	"Blended learning. Offline and online learning."		

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Informant No. 4 and 6	“Online presentations help us in delivering lessons” “Lectures are needed to transfer concepts, theories and ideas to the students.”
Informant No. 3 and 11	“Demonstration and lectures are our strategies to transfer learning” “The focus of our instruction is our students for them to learn independently”. “Students are the center of our teaching”

PERCEIVED EFFECT IN TEACHING PHYSICAL EDUCATION DURING PANDEMIC TO THE PERFORMANCE OF TEACHER

According to the conducted interview to the different teachers who are on the field of Physical Education, Key Informant Number 8 find it really challenging as a P.E Instructor since on this field requires application and performances because the main reason of teaching P.E is to be connected to students and to make them well equipped with knowledge and skills. To some teacher they cannot measure effectiveness as a PE teacher because all of the activities must be done and facilitated physically. Online learning was not only popularized in time of pandemic, even before online class had been the modality in other country most especially who are working on day time. However, this become more challenging to teachers especially in teaching the skills. According to Key Informant Number 6, it always affect the teachers because they use a conducive kind of teaching for students learning, hence, adjustments was not only for students but also for teachers who serves as the instructors. This result conforms with the study of Gutierrez, et al., 2023, which states that the pandemic has had a significant impact on PE teachers, who have faced a multitude of challenges and uncertainties in adapting to new teaching methods and adjustments.

Furthermore, based on the answers of the respondents all of the activities must be physically assisted, some PE teachers find this as a challenging situation to manage specially in integrating knowledge specially skills. Practical implications of the findings suggest that schools and educational institutions should provide adequate training and support for PE teachers to adapt to new teaching and learning modalities during the COVID-19 pandemic or any other untoward circumstances.

TABLE 3. PERCEIVED EFFECT IN TEACHING PHYSICAL EDUCATION DURING PANDEMIC TO THE PERFORMANCE OF TEACHER

Participant	Responses	Clustered Theme	Emergent Theme
Informat No. 3 and 7	“I cannot measure my effectivity as PE teacher because all of	Transfer of learning to skills	➤ Challenging Teachers' Flexibility and

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Informat No. 8	the activities must be done and facilitated physically to the students for them to be aware of the proper mechanics in learning the skills.” “PE in virtual classroom is hard and difficult for me” “For me, is quite challenging but manifesting. In the midst of pandemic. We all know that a challenge for the teachers is how to import knowledge and exhibit the skills of the students when the mode of learning is online. On the other hand, during pandemic both teachers and students become innovative and productive in utilizing in teaching and learning.”	were challenging in the mode of online learning. • Versatile in circumstances • Application and performance of skills are what Physical Educations requires to assess the development of learners’ ability. • Challenging yet fulfilling. • Flexible teaching • The sense of accountability and responsibility. • Become resourceful and creative.	Versatility
Informat No. 2 and 10 Informat No.	“During pandemic its very hard to import knowledge to the students especially introducing the skills to the students. It is very tiring in the part of the teachers because of the students have no materials or sports equipment to be used to enhance their skills” “Pandemic taught me to become a resourceful and flexible teacher” “Giving the teachers		

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9 and 5

an insight how to be resourceful on how to deliver instructions despite of the circumstances brought by pandemic.”

Informat No.
11 and 4

“As PE teacher, I should be flexible in different modalities”
“It was challenging part to every PE teachers to conduct activities face to face, since our field more on application and performances. All we need to do is to keep our student connected and well equipped.”

Informat No.
6 and 1

“My versatile personality have been tested”
“It gives a sense of accountability and responsibility to both students and teachers in dealing the situation as well as the class Online learning wasn't just created, in-just the time of pandemic. The adjustment wasn't only for student but the first attempts are coming from the teacher who serves as the instructors. But well I think, that though it's hard, I see that everyone become more interested in the lit.”

“Teaching PE online is really hard for me”

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(13) Conclusions and Recommendations

CONCLUSIONS

The following conclusions were made based on the study findings, which were described below:

1. The modes of teaching Physical Education during pandemic provides the largest intervention to support the physical activity to maintain quality programming during virtual learning periods remain unexplored. Students are required to comply their activities such as modules, files, pictures, videos and made learning interaction through comments sections, group chat messages and video calls.
2. Teachers utilized approaches, methods, and strategies in teaching physical education for the learners and teachers to cope up the learnings during pandemic. Different teachers used different forms such as online classes with livestreaming, recorded videos, tasks for students, projects or just links to follow.
3. Teaching Physical Education increased time in physical education, a switch from seated activities to more “active” classrooms, and regular physical activity – such as walking, dance, and sports – during the week. Academic performance is affected by several factors: students' learning skills, parental background, peer influence, teachers' quality, and learning infrastructure.

RECOMMENDATIONS

Based on the findings and conclusions of the study, the following recommendations are suggested:

1. Curriculum planner may use this study to design the curriculum for further enhancement in creating an effective program with regards to Physical Education with the use of different strategies applicable to distant learning wherein the compilation could be the basis for the improvement of the curriculum.
2. Through this study teachers will be able to know and help to assess the coping strategies of a class. The compilations will also help the teachers discover and apply new timely and relevant strategies in effective learning for their students. Moreover, it will enable them to reach out effectively for the students of this generation through different modes of learnings with the use of technology and value its importance and significance of today's learning.
3. The results of this study will give the future researchers an idea to study further about the strategies and modes of teaching the Physical Education timely and relevantly. It is recommended for the future researchers to study and replicate to improve and fill the gaps of this study and add different modes of learning that is used and adaptable today.

(14) Output

Maagazine on Teaching Documentary

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and

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